TITANIC SURVIVAL PREDICTION

CODSOFT Data Science Internship

Task 1

Submitted By:

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Introduction

The Titanic Survival Prediction project is a Machine Learning project that predicts whether a passenger survived the Titanic disaster based on passenger information such as age, gender, passenger class, fare, and family details.

This project uses Python libraries for data analysis, preprocessing, visualization, and machine learning model building. Logistic Regression was used to train the prediction model.

The project helps in understanding the complete Machine Learning workflow including data cleaning, visualization, model training, and evaluation.

Objective

The main objective of this project is to build a Machine Learning model that can predict whether a passenger survived the Titanic disaster.

The project aims to:

- Analyze Titanic passenger data
- Perform data cleaning and preprocessing
- Visualize important patterns in the dataset
- Train a predictive Machine Learning model
- Evaluate model performance using accuracy metrics

Dataset Information

The dataset used in this project is the Titanic Dataset obtained from Kaggle.

Dataset Link:

https://www.kaggle.com/datasets/yasserh/titanic-dataset

The dataset contains information about passengers such as:

- **Passenger ID**
- **Name**
- **Age**
- **Gender**
- **Ticket Class**
- **Fare**
- **Cabin**
- **Embarked Port**
- **Survival Status**

Target Variable:

Survived

0 = Did Not Survive

1 = Survived

Technologies Used

The following technologies and libraries were used in this project:

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Tools:

- Jupyter Notebook
- GitHub
- Microsoft Word

Data Preprocessing

Data preprocessing is an important step before training the Machine Learning model.

The following preprocessing steps were performed:

- Checked missing values in the dataset
- Filled missing Age values using median
- Filled missing Embarked values using mode
- Removed Cabin column due to excessive missing values
- Converted categorical columns into numerical values using Label Encoding
- Selected important features for model training

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the relationship between different features and passenger survival.

Several visualizations were created:

- Survival Count Plot
- Gender vs Survival
- Passenger Class vs Survival
- Confusion Matrix

Observations:

- Female passengers had a higher survival rate
- First-class passengers survived more compared to third-class passengers
- Passenger age and fare influenced survival chances.

Model Building

The dataset was divided into training and testing sets using train_test_split().

Machine Learning Model Used:
- Logistic Regression

Training Process:
- Model trained using training dataset
- Predictions generated using testing dataset
- Accuracy score calculated to evaluate model performance

Model Evaluation

The model performance was evaluated using:
- Accuracy Score
- Classification Report
- Confusion Matrix

The Logistic Regression model achieved approximately 80% accuracy on the testing dataset.

The confusion matrix helped visualize correct and incorrect predictions.

Results

The Titanic Survival Prediction model successfully predicted passenger survival with good accuracy.

Key Results:
• Achieved approximately 80% accuracy
• Successfully classified survived and non-survived passengers
• Generated meaningful visualizations for better understanding

The project demonstrates the effectiveness of Machine Learning techniques in predictive analysis.

Conclusion

This project provided practical experience in Machine Learning and Data Science workflows.

The Titanic Survival Prediction model was successfully developed using Logistic Regression. The project improved understanding of:

• Data preprocessing
• Data visualization
• Model training
• Model evaluation

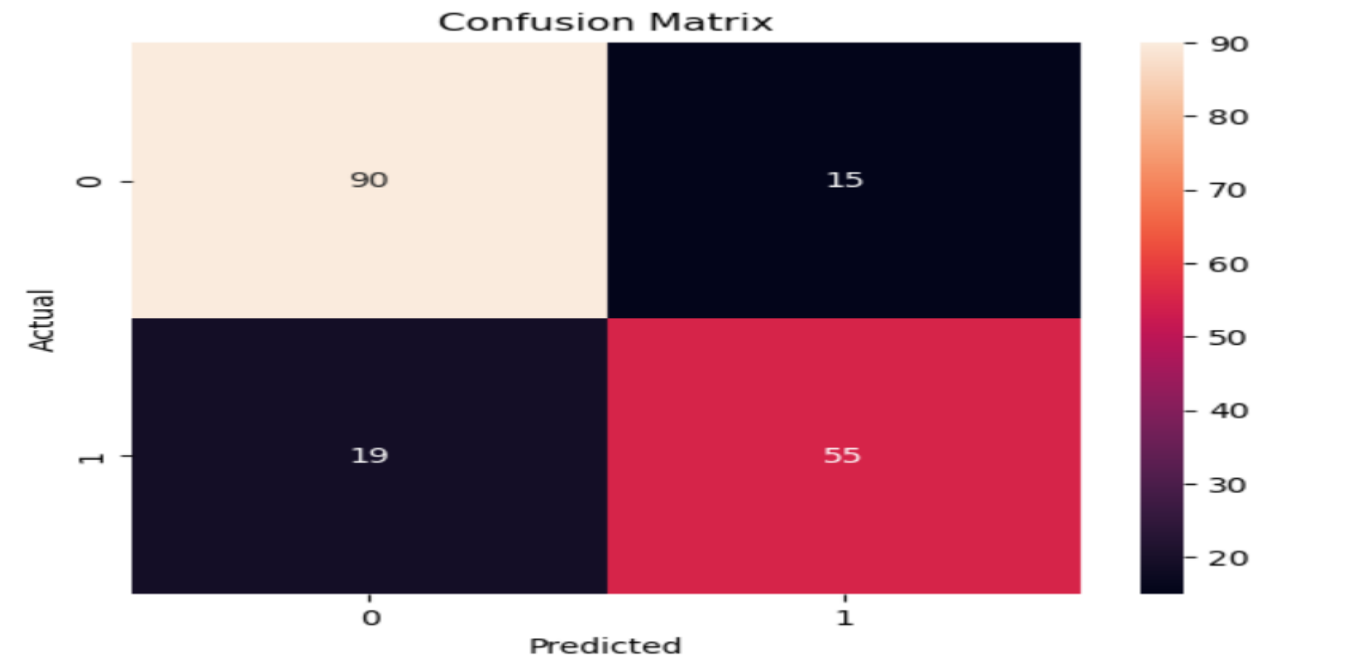
This project also enhanced problem-solving and analytical skills.

Future Improvements

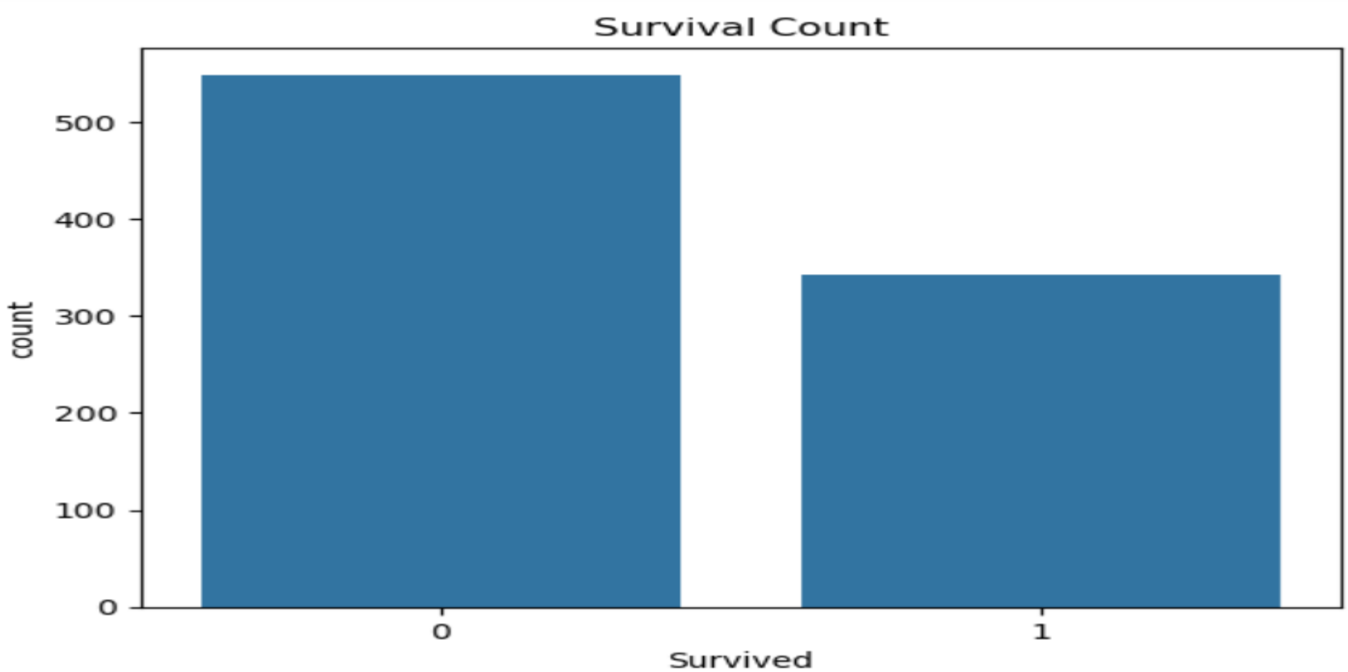
Future improvements for this project include:

• Using advanced Machine Learning algorithms
• Performing feature engineering
• Hyperparameter tuning
• Improving prediction accuracy
• Deploying the model as a web application

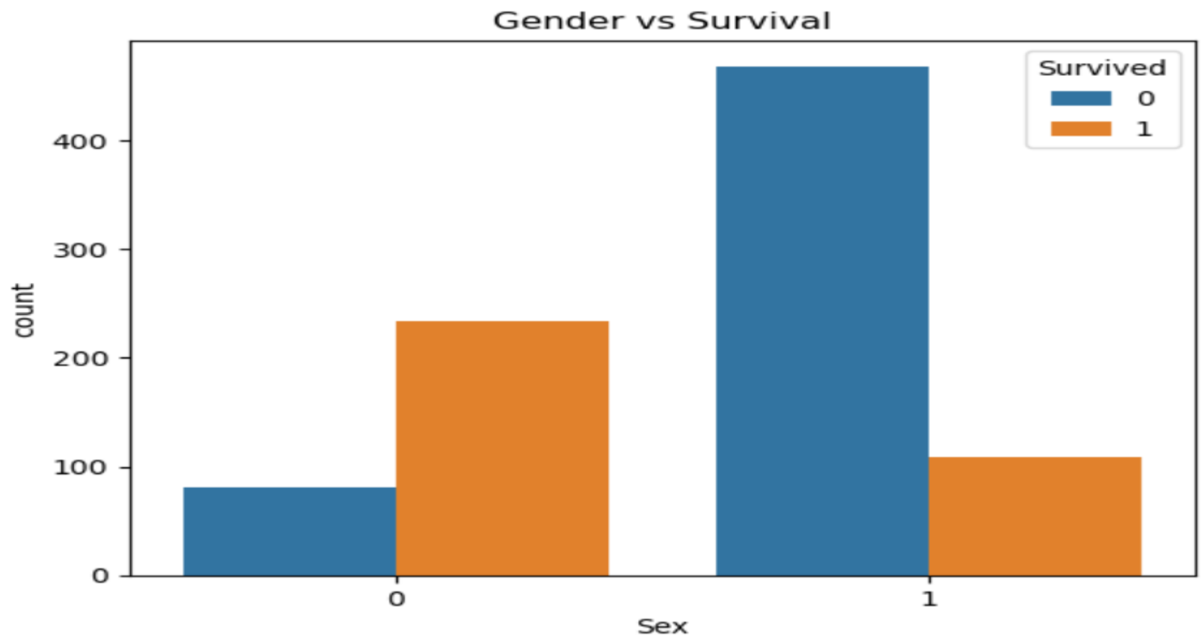
Screenshots



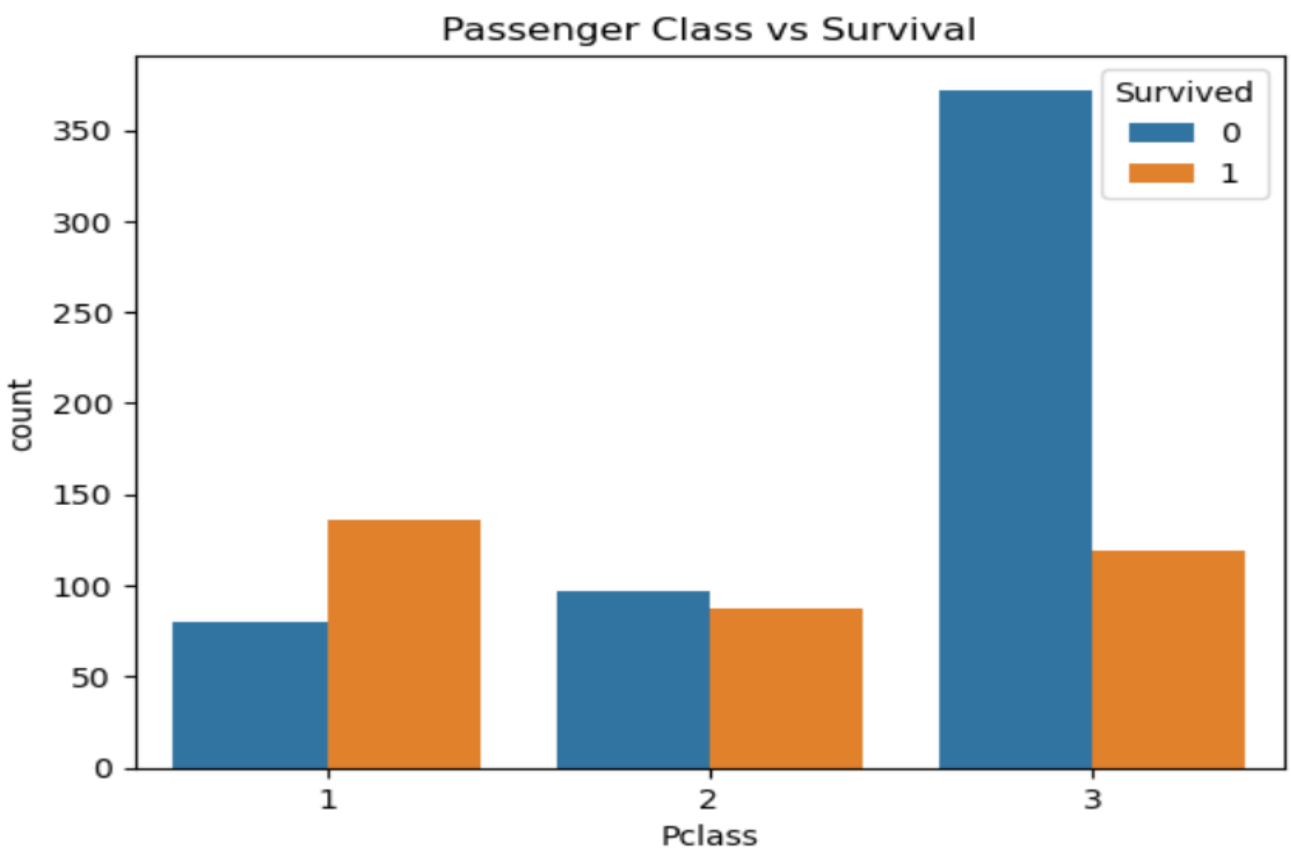
Confusion matrix-Actual vs predicted



Survival count



Gender vs suvival-sex vs count



Passenger class vs survival

GitHub Repository

GitHub Repository Link: https://github.com/Reshma-spec/CODSOFT